DYNAMIC CHEAP TALK WITHOUT FEEDBACK

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Abstract. We study a dynamic sender-receiver game in which the sender observes a state evolving according to a Markov chain but does not observe the receiver's action. Despite the absence of feedback, dynamic interaction partially restores commitment. We show that any equilibrium payoff of a persuasion model with partial commitment—where the sender can deviate only to signaling policies that preserve the marginal distribution over messages—as well as any convex combination of such payoffs across distributions over messages, can be achieved as a uniform equilibrium payoff in the dynamic game. When the sender's payoff is state-independent, she achieves the Bayesian persuasion payoff.

1. Introduction

Communication between an informed expert and an uninformed decision-maker provides a natural framework for studying strategic information transmission. When their interests conflict, the expert may distort messages to steer the outcome in her favor, while the decision-maker acts based on his own interests. This limits the amount of information that can be transmitted and restricts the set of achievable outcomes.

In many such relationships, decisions are made repeatedly over time. This dynamic interaction may help overcome this friction. When uncertainty evolves, information revealed today remains useful but decays over time, creating scope for intertemporal incentives. Such incentives are typically sustained through feedback, with future communication used to reward or punish behavior. This carrot-and-stick argument, however, relies on the expert observing the decision-maker's actions. Yet, in many situations, the expert receives little or no feedback. A natural example is financial advice, where an advisor recommends portfolio allocations to an investor without observing whether they are followed. Another example is a doctor who prescribes a diet plan or treatment to a patient without observing compliance. Can credible communication arise without such feedback? If so, what are its limits?

We consider an infinite-horizon dynamic sender-receiver model in which the state evolves according to a Markov chain. In each period, the sender privately observes the state and sends a message to the receiver, who then chooses an action. The sender gets

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no feedback: she observes neither the receiver's action nor her realized stage payoff. The players' stage payoffs depend only on the current state and the receiver's action.

Despite the absence of feedback, dynamic interaction improves communication. Although the sender does not observe the receiver's actions, the receiver observes the history of messages and can monitor their empirical distribution. This allows him to detect deviations that alter message frequencies, effectively restricting the sender to those that preserve the marginal distribution over messages. As a result, the sender faces endogenous restrictions on her deviations, allowing outcomes beyond those achievable in the one-shot game.

Because players do not observe their stage payoffs, we focus on uniform equilibrium to obtain robust predictions. Uniform equilibrium payoffs are limits of approximate equilibria that are sustained by a single strategy profile for all sufficiently patient players. Our main result (Theorem 1) shows that any equilibrium payoff of a persuasion model with partial commitment can be sustained as a uniform equilibrium payoff in the dynamic game: the receiver best responds to his posterior belief, while the sender can deviate only to signaling policies that preserve the marginal distribution over messages. Moreover, any convex combination of such payoffs across message distributions can also be sustained. This persuasion model is very closely related to Lin and Liu (2024), where the sender's deviations are exogenously restricted to preserve the marginal distribution over messages.

Dynamic communication without feedback partially restores commitment. This improves the sender's payoff relative to the static cheap talk model. In particular, when the sender's payoff is state-independent, she can fully bridge the commitment gap and achieve the Bayesian persuasion payoff, the upper bound in this setting.[1] This contrasts with Kuvalekar, Lipnowski, and Ramos (2022), who characterize the Perfect Bayesian equilibria of a repeated cheap talk game without feedback and show that this upper bound is not attainable.

The intuition is as follows. The game is played in blocks. In each period, the sender induces the same posterior beliefs as in the static model. This is feasible under our assumption of a pseudo-renewal Markov chain. The receiver follows the prescribed response rule as long as message frequencies remain within their quotas. Otherwise, he plays an action corresponding to a different message whose quota has not yet been filled. This ensures that the frequency of each message used matches its quota within each block. For sufficiently long blocks, quotas bind only in a small fraction of periods. As a result, the empirical distribution over states and actions can be made arbitrarily close to the target distribution. The sender's deviations are effectively restricted to those that preserve the marginal distribution over messages, so by the equilibrium conditions of the static model, the induced strategy profile is an approximate equilibrium of the dynamic game. Convex combinations are obtained by varying the target distributions across sub-blocks.

[1] This is the sender's upper bound even with commitment in the dynamic game.

The previous result identifies a tractable subset of uniform equilibrium payoffs. We next examine how tight this characterization is. While the result builds on deviations that preserve the marginal distribution over messages, correlation across periods allows for richer statistical tests, making some such deviations detectable. Example 2 shows that, in general, uniform equilibrium payoffs need not lie in the convex hull of partial commitment payoffs. Nevertheless, Proposition 1 provides a necessary condition: equilibrium outcomes must be robust to a class of undetectable deviations based on permutations of the state process, following the approach of Renault, Solan, and Vieille (2013). These deviations correspond to playing the equilibrium strategy under a relabeled state process that is statistically indistinguishable from the true process and is therefore undetectable. Finally, Proposition 2 shows that when states are i.i.d., all deviations that preserve the distribution over messages are undetectable, as there is no correlation across periods. Hence, the static benchmark—the convex hull of equilibrium payoffs of the persuasion model with partial commitment—is exact. This result provides a micro-foundation for the persuasion model with partial commitment.

Related Literature: Our paper relates to the growing literature on strategic communication in dynamic environments where the underlying state evolves over time, see, e.g., Renault, Solan, and Vieille (2013); Escobar and Toikka (2013); Renault, Solan, and Vieille (2017); Ely (2017); Margaria and Smolin (2018); Meng (2021); Lehrer and Shaiderman (2026). In particular, our paper contributes to the question of whether repeated interaction can substitute for commitment, see, e.g., Kuvalekar, Lipnowski, and Ramos (2022); Best and Quigley (2024); Mathevet, Pearce, and Stacchetti (2024); Jain and Perchet (2026). These papers offer rationales based on coarse summaries of past play, reputation, or statistical tests such as calibration. In contrast, we show that even without feedback, dynamic interaction can partially restore commitment through endogenous restrictions on the distribution over messages.

Our paper is closely related to Kuvalekar, Lipnowski, and Ramos (2022), who also study a repeated cheap talk model without feedback. They consider an i.i.d. environment and characterize Perfect Bayesian equilibrium payoffs via an equivalence to a static cheap talk model with money burning. In contrast, we consider a Markovian setting and study uniform equilibrium payoffs. In their setting, the sender cannot achieve the Bayesian persuasion payoff even when players are very patient. Moreover, when the sender's preferences are state-independent, repetition does not improve her payoff at all, whereas in precisely this case she can achieve the Bayesian persuasion payoff in our setting.

Our equilibrium construction is based on quota mechanisms, where agents are required to match prescribed message frequencies; see, e.g., Jackson and Sonnenschein (2007); Renault, Solan, and Vieille (2013); Escobar and Toikka (2013); Frankel (2016); Ball and Kattwinkel (2025). Our approach is closest to Renault, Solan, and Vieille (2013), where

the receiver monitors message frequencies and replaces them when necessary to meet the prescribed quotas.

Our main result links the dynamic game to a static sender-receiver model, connecting our analysis to the literature on communication with partial commitment. In cheap talk (Crawford and Sobel, 1982; Green and Stokey, 2007), the sender cannot commit, while in Bayesian persuasion (Kamenica and Gentzkow, 2011), she fully commits to a signaling policy. Our static benchmark lies in between: the sender can deviate to any policy that preserves the marginal distribution over messages. Our static model is very closely related to the one in Lin and Liu (2024). One difference is that in their model the sender endogenously chooses the marginal distribution over messages, whereas in our setting it is fixed ex ante.[2] Moreover, in our dynamic environment, players can achieve convex combinations of equilibrium payoffs across distributions over messages. More broadly, our static formulation relates to work that weakens commitment in Bayesian persuasion, see, for instance, Min (2021) and Lipnowski, Ravid, and Shishkin (2022).

The rest of the paper is organized as follows. In Section 2, we introduce the dynamic game without feedback and discuss the value of commitment. In Section 3, we present the persuasion model with partial commitment and provide our main results. In Section 4, we conclude and discuss future work. All omitted proofs are in Appendix A.

2. Model

We consider a dynamic game between two players: the sender (she) and the receiver (he). The state of the world evolves over time in a discrete-time infinite horizon. At each period $n \in \mathbb{N}$, the sender observes the current state $\omega_n \in \Omega$ and sends a message $m_n \in M$ to the receiver. Upon observing the message m_n, the receiver takes an action $a_n \in A$. The action a_n is not disclosed to the sender. The sender's and the receiver's payoff in the period equals $u_S(\omega_n, a_n)$ and $u_R(\omega_n, a_n)$ respectively.[3]

The state evolves according to a Markov chain. Given the current state ω_n, the state in the next period ω_{n+1} is drawn according to the transition matrix $Q(\omega_{n+1} \mid \omega_n)$. We assume the Markov chain is irreducible, aperiodic and pseudo-renewal.

Definition 1. A Markov chain $\{\omega_n\}_{n\in\mathbb{N}}$ is **pseudo-renewal** if $\omega \neq \tilde{\omega}$, then $Q(\omega \mid \tilde{\omega}) = \alpha_\omega$ for some $(\alpha_\omega)_\omega \geq 0$.

A Markov chain is pseudo-renewal if, for any two distinct states, the transition probability depends only on the destination state and not on the current state. This condition

[2]They focus on pure receiver strategies, in which case it is without loss to restrict attention to direct signaling policies.

[3]Neither player observes realized payoffs. In particular, the sender does not observe the action a_n, while the receiver does not observe the state ω_n.

is satisfied, for example, for i.i.d. processes, for binary-state Markov chains, and for shock-driven processes where shocks are i.i.d. and the state remains constant between shocks.[4]

Let $\mu \in \Delta(\Omega)$ denote the unique invariant distribution of the Markov chain.[5] The initial state ω_1 is drawn according to μ. The sets of states Ω, actions A, and messages M are all finite. We assume that $|M| \geq |A|$.

The players' strategies map their information into choices. For the sender, a strategy is a sequence of mappings $\sigma = (\sigma_n)_{n\geq 1}$, where $\sigma_n : (\Omega \times M)^{n-1} \times \Omega \to \Delta(M)$ specifies a distribution over messages given past states and messages and the current state. For the receiver, a strategy is a sequence of mappings $\tau = (\tau_n)_{n\geq 1}$, where $\tau_n : (M \times A)^{n-1} \times M \to \Delta(A)$ specifies a distribution over actions given past messages and actions and the current message. Given a strategy profile (σ, τ) and $\delta \in (0, 1)$, the δ-discounted payoff of player $i \in \{S, R\}$ is

$$\gamma_i^\delta(\sigma, \tau) = \mathbb{E}_{\sigma,\tau}\Big[(1-\delta)\sum_{n=1}^{\infty} \delta^{n-1} u_i(\omega_n, a_n)\Big].$$

Uniform equilibrium: In the dynamic game, players do not observe their stage payoffs, so continuation play cannot be conditioned on realized outcomes. This makes equilibrium behavior sensitive to small perturbations. To obtain robust predictions, we focus on limits of approximate equilibria that are sustained by the same strategy profile for all sufficiently large discount factors. We therefore adopt uniform equilibrium as the solution concept.[6]

Definition 2. A payoff vector $\boldsymbol{\gamma} = (\gamma_S, \gamma_R) \in \mathbb{R}^2$ is a **uniform equilibrium payoff** if for every $\epsilon > 0$, there exist $\delta_0 \in (0, 1)$ and a strategy profile (σ^*, τ^*) such that for all $\delta \geq \delta_0$,

$$\gamma_S^\delta(\tilde{\sigma}, \tau^*) \leq \gamma_S^\delta(\sigma^*, \tau^*) + \epsilon \quad \forall \tilde{\sigma},$$
$$\gamma_R^\delta(\sigma^*, \tilde{\tau}) \leq \gamma_R^\delta(\sigma^*, \tau^*) + \epsilon \quad \forall \tilde{\tau},$$

and

$$\|\boldsymbol{\gamma}^\delta(\sigma^*, \tau^*) - \boldsymbol{\gamma}\|_\infty \leq \epsilon.$$

Thus, a payoff vector is a uniform equilibrium payoff if it is the limit of a sequence of ϵ-equilibrium payoffs, where for each $\epsilon > 0$ a single strategy profile is an ϵ-equilibrium for all sufficiently large discount factors. Let $\mathcal{U}$ denote the set of uniform equilibrium payoffs.

Value of Commitment: We show that the Bayesian persuasion payoff evaluated at the invariant distribution is an upper bound on the sender's equilibrium payoff. To

[4]Renault, Solan, and Vieille (2013, 2017) and Hörner, Mu, and Vieille (2017) also analyze dynamic sender–receiver games in which the Markov chain is pseudo-renewal.

[5]The invariant distribution μ satisfies $\sum_\omega \mu(\omega)Q(\omega' \mid \omega) = \mu(\omega')$ for all $\omega' \in \Omega$.

[6]Uniform equilibrium is a common solution concept in stochastic games, particularly under imperfect monitoring; see, e.g., Lehrer (1990, 1992); Tomala (1999); Vieille (2000); Mertens, Sorin, and Zamir (2015); Deb, González-Díaz, and Renault (2016); Solan (2022).

establish this result, we adopt a belief-based representation of the dynamic game, under which payoffs can be tracked through the receiver's beliefs.

In the dynamic game, the receiver updates his belief over the states given the sender's message. Since the state follows a Markov chain, beliefs evolve across periods. Given any history, let q_t and p_t denote the receiver's belief about the state ω_t before and after observing the message in period t, respectively. The transition matrix Q links the two beliefs:

$$q_{t+1}(\omega) = \sum_{\tilde{\omega} \in \Omega} p_t(\tilde{\omega})\, Q(\omega \mid \tilde{\omega}).$$

Since the sender does not observe the receiver's actions, his choice does not affect future play. Hence, a myopic best response to his posterior belief is optimal in each period. In any ϵ-equilibrium with sufficiently patient players, the receiver therefore follows a myopic best response in all but a small fraction of periods. Consequently, in the limit, the discounted payoff coincides with that obtained when the receiver plays a myopic best response in every period and can therefore be tracked using the induced belief process.

For $i \in \{S, R\}$, define the indirect utility

$$\hat{u}_i(p) := \mathbb{E}_p[u_i(\omega, a^*(p))],$$

where $a^*(p)$ is the receiver's best response given belief p.[7]

In any period, the sender's strategy induces a Bayes-plausible distribution over posterior beliefs, i.e., the expected posterior equals the prior. In the static persuasion model, Kamenica and Gentzkow (2011) show that the sender's maximal payoff, given prior π, is the concave envelope of the sender's indirect utility, denoted by $\operatorname{Cav} \hat{u}_S(\pi)$.

The same reasoning applies period by period in the dynamic game. In period n, the sender induces a Bayes-plausible distribution over posterior beliefs with prior q_n, so her expected payoff is at most $\operatorname{Cav} \hat{u}_S(q_n)$. Therefore, the sender's equilibrium discounted payoff satisfies

$$\begin{aligned}
\gamma_S^\delta &\leq (1-\delta) \sum_{n=1}^{\infty} \delta^{n-1} \mathbb{E}_\sigma[\operatorname{Cav} \hat{u}_S(q_n)] \\
&\leq (1-\delta) \sum_{n=1}^{\infty} \delta^{n-1} \operatorname{Cav} \hat{u}_S(\mathbb{E}_\sigma[q_n]) \\
&= \operatorname{Cav} \hat{u}_S(\mu),
\end{aligned}$$

where the second inequality follows from Jensen's inequality, and the last equality uses $\mathbb{E}_\sigma[q_n] = \mu$ for all n. Thus, the Bayesian persuasion payoff at μ is an upper bound in the dynamic game.[8]

[7] If multiple actions are optimal, we assume ties are broken in favor of the sender, as is standard in the persuasion literature.

[8] Renault, Solan, and Vieille (2017) obtain the same upper bound even when the sender can commit against short-run receivers.

The stage game corresponds to cheap talk, where the sender cannot commit to how messages are generated.[9] When the sender's preferences are state-independent, Lipnowski and Ravid (2020) show that her maximal payoff is given by the quasi-concave envelope of $\hat{u}_S$. The gap between the concave and quasi-concave envelopes captures the value of commitment (see Figure 1a).

Our objective is to characterize how much of this gap can be recovered through dynamic interaction without feedback. We show that the sender's equilibrium payoff improves upon cheap talk, partially restoring commitment. In particular, when the sender's preferences are state-independent, she attains the Bayesian persuasion payoff.

3. Main Results

Our main result partially characterizes the set of uniform equilibrium payoffs in terms of a static persuasion model with partial commitment. We first define the static model. We then show that any equilibrium payoff of the static model, and its convexification, can be sustained as a uniform equilibrium payoff. Finally, we show that there may exist equilibrium payoffs outside this set and provide a necessary condition for equilibrium.

3.1. Persuasion with partial commitment.

The key benchmark in our analysis is a static persuasion model with partial commitment. The environment coincides with the stage game of the dynamic model: the sets of states Ω, messages M, and actions A, as well as payoff functions u_S and u_R, are the same.

Fix a prior $\pi \in \Delta(\Omega)$ and a distribution over messages $\lambda \in \Delta(M)$. The sender chooses a signaling policy $\rho : \Omega \to \Delta(M)$ satisfying

$$\sum_{\omega \in \Omega} \pi(\omega)\rho(m \mid \omega) = \lambda(m) \quad \text{for all } m \in M,$$

and the receiver takes an action after observing the message. An equilibrium requires that the receiver best responds to posterior beliefs and that the sender cannot profitably deviate within the set of policies that preserve the marginal distribution λ. A babbling equilibrium always exists, in which messages are uninformative and the receiver best responds to the prior belief.

The interpretation is that the receiver can detect any change in the distribution of messages, so the sender is effectively restricted to policies that preserve this distribution. Such constraints may arise from institutional requirements, for instance, grading policies that require instructors to match a fixed grade distribution set by the university.

We formulate equilibrium directly in terms of the posterior beliefs $\{p_m\}_{m \in M}$. A feasible signaling policy induces posterior beliefs $\{p_m\}_{m \in M} \in \Sigma(\pi, \lambda)$.

[9] Any cheap talk equilibrium payoff under the invariant distribution μ is achievable as a uniform equilibrium payoff.

Equivalently, any such collection induces a joint distribution $\nu \in \Delta(\Omega \times A)$ given by $\nu(\omega, a) = \sum_{m \in M} \lambda(m) p_m(\omega) \kappa(a \mid m)$. The expected payoff of player $i \in \{S, R\}$ is

$$\sum_{m \in M} \lambda(m) \sum_{\omega \in \Omega} p_m(\omega) \sum_{a \in A} \kappa(a \mid m) u_i(\omega, a).$$

We formulate equilibrium directly in terms of the posterior beliefs $\{p_m\}_{m \in M}$ and receiver's response rule $\kappa : M \to \Delta(A)$.

Receiver. The receiver's incentive constraint reduces to best response: for every $m \in M$ with $\lambda(m) > 0$,

$$\text{(Eq-R)} \qquad \kappa(\cdot \mid m) \in A^*(p_m),$$

where $A^*(p) = \arg\max_{b \in A} \sum_\omega p(\omega) u_R(\omega, b)$ denote the receiver's best responses given belief p.

Sender. The sender cannot profitably deviate among all Bayes-plausible posterior collections with weights λ. That is, for any alternative $\{\tilde{p}_m\}_{m \in M} \in \Sigma(\pi, \lambda)$,

$$\text{(Eq-S)} \qquad \sum_{m \in M} \lambda(m) \sum_{\omega} \Big(p_m(\omega) - \tilde{p}_m(\omega) \Big) \sum_{a \in A} \kappa(a \mid m) u_S(\omega, a) \geq 0.$$

A collection of posterior beliefs $\{p_m\}_{m \in M}$ and response rule $\kappa : M \to \Delta(A)$ is an equilibrium if it satisfies Bayes plausibility, the best response condition (Eq-R), and the condition that the sender has no profitable undetectable deviation (Eq-S). Let $\mathcal{E}(\pi, \lambda)$ denote the set of equilibrium payoffs given prior π and marginal λ.

To illustrate the persuasion model with partial commitment, consider the following example with binary messages, adapted from Lipnowski and Ravid (2020).

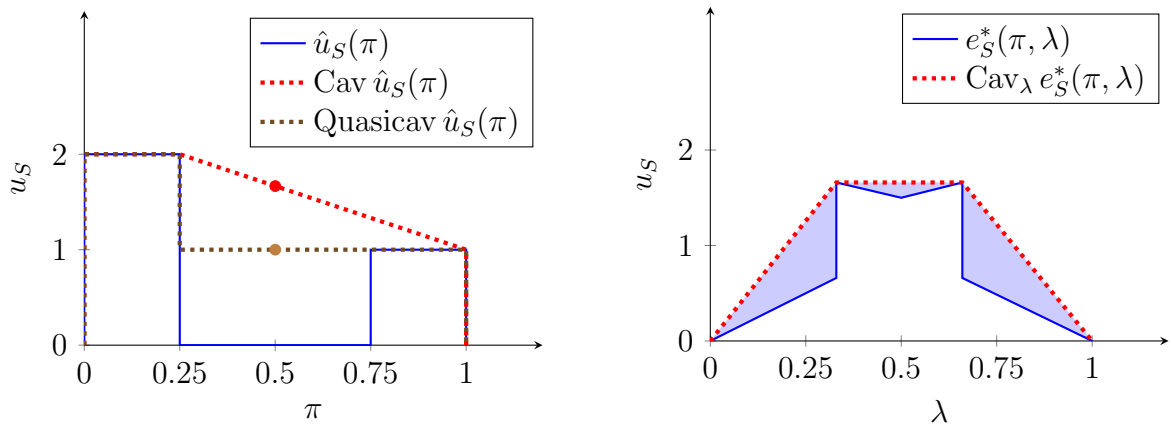
Example 1. *A political think tank advises a lawmaker on whether to implement one of two reforms, denoted* 1 *and* 2*, or to implement no reform. The corresponding action set is* $A = \{a_1, a_2, a_0\}$*. The think tank's payoff is state-independent: it prefers reform 2 to reform 1, and any reform to no reform. The lawmaker wants to implement a reform only if sufficiently confident that it is good. Let* $\Omega = \{\omega_1, \omega_2\}$*, where* ω_i *is the state in which reform* i *is good, and let* $M = \{m_1, m_2\}$ *denote the message set. The think tank and the lawmaker's preferences are captured by the following payoff matrix:*

	a_2	a_1	a_0
ω_1	$(2, 0)$	$(1, 4)$	$(0, 3)$
ω_2	$(2, 4)$	$(1, 0)$	$(0, 3)$

The lawmaker's optimal action, as a function of $p = \mathbb{P}(\omega_1)$*, is*

$$a^*(p) = \begin{cases} a_2 & \text{if } p \leq \frac{1}{4}, \\ a_0 & \text{if } \frac{1}{4} \leq p \leq \frac{3}{4}, \\ a_1 & \text{if } p \geq \frac{3}{4}. \end{cases}$$

Under Bayesian persuasion and cheap talk, the think tank's maximal payoff is given by the concave and quasi-concave envelopes of the indirect utility evaluated at the prior (see Figure 1a). Their difference captures the value of commitment.



(a) Bayesian persuasion vs cheap talk

(b) Partial commitment payoff for $\pi = 0.5$

FIGURE 1. Comparison of benchmarks and partial commitment

Consider persuasion with partial commitment. The marginal λ restricts the set of feasible posterior beliefs. For instance, let $\pi = (\frac{1}{2}, \frac{1}{2})$ and $\lambda = (\frac{1}{2}, \frac{1}{2})$. Then any feasible pair of posteriors must satisfy $p_{m_1}(\omega_1) = \frac{1}{2} + \epsilon$ and $p_{m_2}(\omega_1) = \frac{1}{2} - \epsilon$ for some $\epsilon \in [-\frac{1}{2}, \frac{1}{2}]$. The think tank's optimal strategy is to induce $p_{m_1}(\omega_1) = 1$ and $p_{m_2}(\omega_1) = 0$, following which the receiver takes actions a_1 and a_2, respectively. To see that this constitutes an equilibrium, note that the receiver best responds to these beliefs, and any deviation by the sender that preserves λ does not affect her payoff. The sender's expected payoff is $\frac{3}{2}$, which is strictly below the Bayesian persuasion payoff of $\frac{5}{3}$.

Figure 1b illustrates the think tank's maximal equilibrium payoff $e_S^*(\pi, \lambda)$ as a function of $\lambda = \lambda(m_1)$ (blue solid line). In particular, when $\lambda = \frac{1}{3}$, the think tank attains the Bayesian persuasion payoff. Convex combinations across marginals yield the concave envelope $\mathrm{Cav}_\lambda\, e_S^*(\pi, \lambda)$ (red dotted line). In the dynamic game, the think tank can achieve all such convex combinations (shaded region). This shows that allowing for variation across message distributions expands the set of attainable payoffs and can improve the sender's payoff relative to any fixed marginal.

This model lies between Bayesian persuasion and cheap talk. In Bayesian persuasion, the sender faces no restriction on the marginal distribution over messages and cannot deviate after choosing a policy, whereas in cheap talk the sender can deviate freely. Here, deviations are restricted to policies that preserve the marginal distribution over messages. This formulation is closely related to Lin and Liu (2024), except that we fix the marginal distribution λ ex ante rather than allowing the sender to endogenously choose it.

Let $e^*(\pi, \lambda) \in \mathcal{E}(\pi, \lambda) \subset \mathbb{R}^2$ denote a sender-optimal equilibrium payoff vector, and define

$$e_S^*(\pi) := \max_{\lambda \in \Delta(M)} e_S^*(\pi, \lambda).$$

Remark 1. Any cheap talk equilibrium is also an equilibrium of the persuasion model with partial commitment with the induced marginal distribution of messages.

This follows since the receiver's incentive constraint is unchanged, while cheap talk allows a larger set of deviations than partial commitment.

Let $u_S^{CT}(\pi)$ and $u_S^{BP}(\pi) = \operatorname{Cav} \hat{u}_S(\pi)$ denote the sender's maximal equilibrium payoff in cheap talk and Bayesian persuasion, respectively. We have

$$u_S^{CT}(\pi) \leq e_S^*(\pi) \leq u_S^{BP}(\pi).$$

Remark 2. If the sender's payoff is state-independent, then $e_S^*(\pi) = u_S^{BP}(\pi)$.

When the sender's payoff is state-independent, deviations that preserve the distribution over messages do not change the induced distribution over actions, and hence do not affect the sender's payoff. Thus, the sender attains the Bayesian persuasion payoff.

3.2. **Uniform equilibrium payoffs.** In this section, we show that any equilibrium payoff of the static persuasion model with partial commitment and any convex combination of such payoffs, can be sustained as a uniform equilibrium payoff. The characterization depends only on the invariant distribution of the transition matrix.

We now state the main result. It establishes that any equilibrium payoff of the static persuasion model with partial commitment, with prior given by the invariant distribution, as well as any convex combination thereof, can be sustained as a uniform equilibrium payoff in the dynamic game. Given prior μ, let $\operatorname{Co}_\lambda\big(\mathcal{E}(\mu, \lambda)\big)$ denote the set of convex combinations of equilibrium payoffs across distributions over messages.[10]

Theorem 1. *Any equilibrium payoff of the persuasion model with partial commitment and prior μ, as well as any convex combination of such payoffs across distributions over messages, is a uniform equilibrium payoff. Formally,*

$$\operatorname{Co}_\lambda\big(\mathcal{E}(\mu, \lambda)\big) \subseteq \mathcal{U}.$$

Proof Sketch: We first show that any equilibrium payoff $e(\mu, \lambda)$ of the persuasion model with partial commitment is a uniform equilibrium payoff. We then show that any convex combination of such payoffs is also a uniform equilibrium payoff.

Fix $\lambda \in \Delta(M)$ and an equilibrium payoff $e(\mu, \lambda)$. Let $\{p_m\}_{m \in M}$ denote the equilibrium posterior beliefs, which satisfy Bayes plausibility under weights λ and let $\kappa : M \to \Delta(A)$ denote the receiver's best response rule.

[10] $\operatorname{Co}_\lambda\big(\mathcal{E}(\mu, \lambda)\big) = \{\sum_{i=1}^n \eta_i\, e(\mu, \lambda_i) \mid \lambda_i \in \Delta(M),\ \eta_i \geq 0,\ \sum_{i=1}^n \eta_i = 1\}$.

We construct a strategy profile (σ^*, τ^*). The play is divided into consecutive blocks of length N.[11] At the start of each block, players discard past play and restart playing a new block. For any rational λ, choose N such that $N\lambda(m)$ is an integer for all $m \in M$.

Sender's strategy: The sender mimics in every period the same posterior beliefs $\{p_m\}_{m\in M}$ as in the equilibrium of the persuasion model with partial commitment, so the induced joint distribution over states and messages matches that equilibrium outcome.

For each $m \in M$, let $q_m = p_m Q$ denote the next-period belief following message m. Under the pseudo-renewal Markov chain, we have

$$q_m = \alpha p_m + (1-\alpha) \sum_{\tilde{m}\in M} \lambda(\tilde{m}) p_{\tilde{m}} = \alpha p_m + (1-\alpha)\mu,$$

so q_m lies on the line segment joining p_m and μ.[12]

Using this decomposition, the sender chooses the current message as a function of the current state and the previous message:[13]

$$\sigma_n^*(m_n = m \mid \omega_n = \omega, m_{n-1} = m') := \frac{p_m(\omega)}{q_{m'}(\omega)} \Big((1-\alpha)\lambda(m) + \alpha \mathbf{1}_{\{m=m'\}} \Big).$$

The evolution of beliefs induced by the strategy is illustrated in Figure 2.

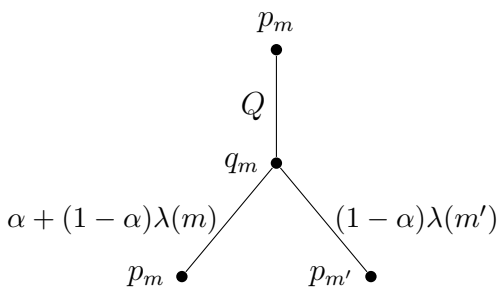


FIGURE 2. Recursive decomposition of receiver's beliefs under σ^*

Receiver's strategy: In each period n, the sender sends a message m_n. If the message has not yet reached its quota, the receiver plays according to $\kappa(\cdot \mid m_n)$, which is optimal given his posterior belief. Otherwise, he ignores the message and instead plays an action corresponding to a different message m' whose quota is not yet exhausted. Note that τ^* is not a sequential best response at all histories, since enforcing quotas may require the receiver to take actions that are not best responses given his posterior belief.[14]

[11] Block strategies are commonly used in the analysis of uniform equilibria under imperfect monitoring to detect deviations via statistical tests; see, for example, Lehrer (1990, 1992); Fudenberg and Levine (1991); Tomala (1999); Deb, González-Díaz, and Renault (2016).

[12] For the construction, it suffices that posterior beliefs are closed under the Markov update, i.e., $p_m Q \in \mathrm{Co}\{p_r\}_{r\in M}$ for all m. This condition is closely linked to M-absorbing sets in Lehrer and Shaiderman (2026).

[13] For $n = 1$, the sender uses the equilibrium strategy of the static model with partial commitment.

[14] This contrasts with Kuvalekar, Lipnowski, and Ramos (2022), who focus on Perfect Bayesian equilibria.

To track message frequencies, let θ_n denote the message used by the receiver in period n, and define

$$C_n(m) := \sum_{k=1}^{n} \mathbf{1}_{\{\theta_k = m\}}$$

as the number of times message m has been used up to period n.

According to τ^*, if $C_{n-1}(m_n) < N\lambda(m_n)$, then the receiver plays according to the message sent, i.e., $\theta_n = m_n$. Otherwise, he chooses a message $m' \neq m_n$ such that $C_{n-1}(m') < N\lambda(m')$. For example, m' can be chosen randomly with probability proportional to the remaining quota $N\lambda(m') - C_{n-1}(m')$. Hence, in each block, each message m is used exactly $N\lambda(m)$ times.

The proof proceeds in two steps. First, we show that the payoff induced by (σ^*, τ^*) can be made arbitrarily close to $e(\mu, \lambda)$ for sufficiently patient players. Second, we show that, for any $\epsilon > 0$, the profile (σ^*, τ^*) is an ϵ-equilibrium for sufficiently patient players.[15]

Under (σ^*, τ^*), the expected joint distribution over states and messages in any period n coincides with the static distribution, i.e.,

$$\mathbb{E}_{\sigma^*,\tau^*}\left[\mathbf{1}_{\{\omega_n=\omega, m_n=m\}}\right] = \lambda(m)p_m(\omega).$$

Hence, for a large block N, the receiver acts according to the sender's message in most periods with high probability. Therefore, by choosing N large enough and the players sufficiently patient, the induced payoff can be made arbitrarily close to $e(\mu, \lambda)$.

Now, we show that for any $\epsilon > 0$, the strategy profile (σ^*, τ^*) is an ϵ-equilibrium for all sufficiently patient players.

First, consider the receiver's deviations. Since the sender does not observe the receiver's actions, his current choice does not affect future play. Hence, a myopic best response in each period is optimal. Under τ^*, whenever the quota is not exhausted, the receiver responds to the message sent m_n according to $\kappa(\cdot \mid m_n)$, which is a best response given his posterior belief. For a sufficiently large block N, this occurs in all but a small fraction of periods. Therefore, τ^* coincides with a myopic best response in all but a small fraction of periods, so any gain from deviation can be made arbitrarily small for sufficiently patient players.

Next, consider the sender's deviations. The receiver's strategy enforces that the frequency of message m used in each block equals $N\lambda(m)$. Hence, any sender deviation $\tilde{\sigma}$ induces, within a block, a collection of conditional distributions over states $\{\tilde{p}_m\}_{m\in M}$, where for each m with $\lambda(m) > 0$,

$$\tilde{p}_m(\omega) := \frac{1}{N\lambda(m)}\mathbb{E}_{\tilde{\sigma},\tau^*}\left[\sum_{n=1}^{N} \mathbf{1}_{\{\omega_n=\omega,\, \theta_n=m\}}\right].$$

[15] The profile (σ^*, τ^*) is also a uniform equilibrium with full feedback. This relates to Proposition 2 in Renault, Solan, and Vieille (2013), which shows that Nash equilibrium payoffs without feedback are a subset of those with feedback.

These distributions satisfy Bayes plausibility with prior μ and weights λ, i.e., $\mu = \sum_m \lambda(m)\tilde{p}_m$.

This corresponds to a feasible deviation in the static model with marginal λ. By optimality of $\{p_m\}_{m\in M}$, such deviations cannot yield a higher payoff. Moreover, for sufficiently large N and sufficiently patient players, the sender's discounted payoff from any deviation is arbitrarily close to the payoff induced by the corresponding static deviation. Hence, no profitable deviation exists. Therefore, $e(\mu, \lambda) \in \mathcal{U}$.

We now extend the previous block construction to obtain convex combinations of equilibrium payoffs. Let $\lambda = \sum_{i=1}^{n} \eta_i \lambda_i$ with $\eta_i \geq 0$ and $\sum_i \eta_i = 1$. The players still use a block strategy. However, each block of length N is further divided into sub-blocks of lengths $\{N\eta_i\}_{i\in I}$, and in sub-block i the players play according to the strategy profile associated with λ_i.

Using the same argument as before, each sub-block yields an average payoff arbitrarily close to $e(\mu, \lambda_i)$ for sufficiently large N and sufficiently patient players. Since sub-block i occupies a fraction η_i of each block, the overall payoff is arbitrarily close to $\sum_{i=1}^{n} \eta_i e(\mu, \lambda_i)$. Moreover, because the strategy restarts at the beginning of each sub-block, deviations can be evaluated separately within each sub-block, and the same incentive argument applies.

3.3. **On the Converse.** Theorem 1 establishes that all payoff vectors that can be obtained as convex combinations of equilibrium payoffs of the persuasion model with partial commitment are uniform equilibrium payoffs. A natural question is whether the converse inclusion also holds.

A direct extension of the static benchmark would suggest that any outcome that admits a profitable deviation in the set of feasible Bayes-plausible posterior vectors, holding fixed the long-run frequency of messages, should fail to be an equilibrium. However, in the dynamic environment, not all such deviations are feasible. Moreover, the receiver may condition on richer statistical tests, such as frequencies of pairs or longer strings of messages.

The following example shows that some uniform equilibrium outcomes lie outside the convex hull of equilibrium outcomes of the static persuasion model with partial commitment.

Example 2. *Consider binary states $\Omega = \{\omega_1, \omega_2\}$ and let $M \equiv A = \{a_1, a_2, a_3, a_4\}$, where messages are interpreted as action recommendations. Since the state space is binary, we identify a belief with the probability $p = \mathbb{P}(\omega_2) \in [0, 1]$. The invariant distribution is $\mu = \frac{1}{2}$ and is induced by the Markov chain*

$$Q = \begin{bmatrix} \frac{2}{3} & \frac{1}{3} \\ \frac{1}{3} & \frac{2}{3} \end{bmatrix}.$$

Let the posteriors be $p_{a_1} = 0$, $p_{a_2} = \frac{1}{3}$, $p_{a_3} = \frac{2}{3}$, and $p_{a_4} = 1$, with $\lambda = (\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4})$. These satisfy Bayes plausibility: $\sum_a \lambda(a)p_a = \frac{1}{2}$,. The belief dynamics are recursive: $p_{a_1}Q = p_{a_2}$ and $p_{a_4}Q = p_{a_3}$.

The sender's and receiver's payoffs are given by

	a_1	a_2	a_3	a_4
ω_1	$(1,8)$	$(-2,7)$	$(3,3)$	$(-1,0)$
ω_2	$(-1,0)$	$(3,3)$	$(-2,7)$	$(1,8)$

The receiver's optimal action is a_1 for $p \in [0, \frac{1}{4}]$, a_2 for $p \in (\frac{1}{4}, \frac{1}{2}]$, a_3 for $p \in [\frac{1}{2}, \frac{3}{4})$, and a_4 for $p \in [\frac{3}{4}, 1]$ (with ties broken in favor of the sender). Accordingly, it is optimal for the receiver to play action a_i after each message a_i where $i \in \{1, 2, 3, 4\}$.

In every odd period, the sender sends a_1 or a_4, each with probability $1/2$. In every even period, she sends a_2 after a_1 and a_3 after a_4. The receiver follows the recommendation as long as only the pairs (a_1, a_2) and (a_4, a_3) occur; if (a_1, a_3) or (a_4, a_2) is observed, he ignores all future recommendations and plays his optimal action at the prior forever.

Along the prescribed path, the sender is truthful in odd periods. In even periods, however, she prefers to swap the continuation rule: at $p_{a_2} = \frac{1}{3}$ she prefers a_3 to a_2, while at $p_{a_3} = \frac{2}{3}$ she prefers a_2 to a_3. Thus, she has a profitable deviation in the static persuasion model with fixed marginal λ, namely to send a_3 after a_1 and a_2 after a_4, which preserves the one-period distribution.

In the dynamic game, this deviation is not feasible, since it changes the pattern $a_1 \to a_2$, $a_4 \to a_3$ into $a_1 \to a_3$, $a_4 \to a_2$, thereby altering the distribution over adjacent recommendation pairs. It is therefore detected and triggers the babbling outcome.[16]

Hence, the induced payoff vector u belongs to the set $\mathcal{U}$ of uniform equilibrium payoffs. By Proposition 1, every payoff vector in $\mathrm{Co}_\lambda\big(\mathcal{E}(\mu, \lambda)\big)$ arises from a Bayes-plausible splitting of the prior and can be written as a convex combination of payoff vectors induced by posterior beliefs. Since the sender's indirect utility is minimized at the prior π, where it equals $\frac{1}{2}$, every payoff vector in the static benchmark gives the sender at least $\frac{1}{2}$. Because $u_S = \frac{1}{3}$, it follows that $u \notin \mathrm{Co}_\lambda\big(\mathcal{E}(\mu, \lambda)\big)$.

This example shows that dynamically feasible deviations may be strictly smaller than those that preserve quotas. We therefore focus on a class of deviations that are always feasible and undetectable, yielding a necessary condition.

Consider any outcome $\nu \in \Delta(\Omega \times A)$ with prior $\pi \in \Delta(\Omega)$. Let

$$\mathcal{C}(\pi) := \left\{ c \in \Delta(\Omega \times \Omega) : \sum_{\omega'} c(\omega, \omega') = \pi(\omega) \text{ and } \sum_{\omega} c(\omega, \omega') = \pi(\omega') \right\}$$

denote the set of copulas with marginals π.

[16] A deviation that swaps the odd-period recommendations a_1 and a_4 while preserving the continuation pattern is feasible but not profitable. Such a deviation yields an average payoff of $\frac{-1+\frac{4}{3}}{2} = \frac{1}{6}$, which is strictly below the equilibrium payoff $\frac{1}{3}$.

Define the set of state permutations

$$\mathcal{M}(\nu) := \left\{ \tilde{\nu} \in \Delta(\Omega \times A) : \tilde{\nu}(\omega, a) = \sum_{\omega'} c(\omega, \omega')\, \nu(\omega', a) \text{ for some } c \in \mathcal{C}(\pi) \right\}.$$

Definition 3. An outcome $\nu \in \Delta(\Omega \times A)$ with marginal π over states is **robust to state permutations** if

$$\mathbb{E}_\nu[u_S] \geq \mathbb{E}_{\tilde{\nu}}[u_S] \quad \text{for every } \tilde{\nu} \in \mathcal{M}(\nu).$$

State permutations form a subset of deviations that preserve the marginal distribution over messages. They correspond to relabeling states via a permutation and applying the equilibrium strategy to the resulting fictitious process. As shown in Renault, Solan, and Vieille (2013), in pseudo-renewal Markov chains such deviations preserve the law of the state process and are therefore undetectable.

Proposition 1. *If an outcome $\nu \in \Delta(\Omega \times A)$ is induced by a uniform equilibrium, then ν satisfies obedience and is robust to state permutations. Moreover, its marginal over states equals the invariant distribution μ.*

Proposition 1 is not sufficient in general, since $\mathcal{M}(\nu)$ may be strictly smaller than the set of deviations in $\Sigma(\pi, \lambda)$[17] An immediate implication is that every uniform equilibrium outcome is a Bayes correlated equilibrium (Bergemann and Morris, 2016) with prior μ, since, in the limit, each period induces a Bayes plausible distribution of beliefs with prior μ that satisfies obedience.

In Renault, Solan, and Vieille (2013), state permutations are both necessary and sufficient to characterize equilibrium outcomes. In our setting, however, the sender sends messages rather than reporting states, so deviations that preserve frequencies may still alter the temporal structure of messages and become detectable. Thus, while permutation deviations remain necessary, they are not sufficient to characterize equilibrium outcomes, as illustrated in Example 2.

Finally, we consider the special case where the states are drawn i.i.d. with distribution μ. In this case, the dynamic restrictions discussed above disappear, and the static partial-commitment benchmark becomes exact.

Proposition 2. *Suppose that the state process is i.i.d. with distribution μ. Then the set of uniform equilibrium payoffs coincides with the convex hull of static persuasion payoffs with partial commitment:*

$$\mathcal{U} = \mathrm{Co}_\lambda\big(\mathcal{E}(\mu, \lambda)\big).$$

The inclusion follows from Theorem 1. For the converse, each period induces a splitting of μ into posterior beliefs with weights λ. If this splitting does not belong to $\mathcal{E}(\mu, \lambda)$, then

[17] To see this, suppose $\rho(a \mid \omega) > 0$ for all ω, a, so that each posterior p_m has full support. State permutations preserve support and therefore cannot generate degenerate posteriors, whereas such posteriors are feasible in $\Sigma(\pi, \lambda)$.

the sender has a profitable undetectable deviation preserving λ. Since states are drawn i.i.d. across periods, this deviation can be implemented independently across periods, yielding a contradiction. Hence, the induced payoff must belong to $\mathcal{E}(\mu, \lambda)$, and the overall payoff is a convex combination of such payoffs across message distributions.

4. Discussion and Conclusion

We studied a dynamic sender-receiver game, where the sender gets no feedback on the receiver's action. We partially characterize the set of uniform equilibrium payoffs by relating the dynamic game to a static persuasion model with partial commitment. Crucially, dynamic interaction helps the sender achieve a higher payoff despite no feedback. In particular, if the sender's payoff is state-independent, she is able to achieve the Bayesian persuasion payoff.

Many problems remain open in this setting. First, fully characterizing the set of all uniform equilibrium payoffs remains an open problem. Second, exploring scenarios in which the sender receives imperfect feedback on the receiver's actions presents another avenue for investigation. Third, it would be interesting to study the role of more complex statistical tests by the receiver, such as tests based on higher-order frequencies or longer histories of messages (see Escobar and Toikka, 2013; Renou and Tomala, 2015; Hörner, Mu, and Vieille, 2017), and how these affect the set of equilibrium outcomes. Additionally, extending the analysis to more general dynamic environments represents a promising direction.

Appendix A. Omitted Proofs

A.1. **Proof of Theorem 1.** First, we prove that any equilibrium payoff of the static persuasion model with partial commitment is a uniform equilibrium payoff. Then we show any convex combination of these payoffs with respect to the distributions over messages is also a uniform equilibrium payoff.

Fix $e(\mu, \lambda) \in \mathcal{E}(\mu, \lambda)$. Let (σ^*, τ^*) denote the strategy profile constructed in the proof sketch. The proof proceeds in two steps. First, we show that the payoff induced by (σ^*, τ^*) can be made arbitrarily close to $e(\mu, \lambda)$ for sufficiently patient players. Second, we show that for any $\epsilon > 0$, the strategy profile (σ^*, τ^*) is an ϵ-equilibrium for all sufficiently patient players.

Lemma 1. *For every $\epsilon > 0$, there exists $N_0 \in \mathbb{N}$ and $\delta_0 < 1$ such that, for every $N \geq N_0$ and $\delta \geq \delta_0$, the strategy profile (σ^*, τ^*) induces a payoff within ϵ of $e(\mu, \lambda)$.*

Proof. First, we show that at any period n, the expected joint distribution over state ω and the message m equals $\lambda(m)p_m(\omega)$.

$$\begin{aligned}\mathbb{E}_{\sigma^*}[\mathbf{1}_{\{\omega_n=\omega,m_n=m\}}] &= \sum_{\tilde{m}} \sigma_n^*(m_n = m \mid m_{n-1} = \tilde{m}, \omega)\mathbb{P}_{\sigma^*,\tau^*}(m_{n-1} = \tilde{m}, \omega_n = \omega),\\ &= \sum_{\tilde{m}} \frac{p_m(\omega)}{q_{\tilde{m}}(\omega)}\big((1-\alpha)\lambda(m) + \alpha\mathbf{1}_{\{\tilde{m}=m\}}\big)q_{\tilde{m}}(\omega)\lambda(\tilde{m}),\\ &= \lambda(m)p_m(\omega).\end{aligned}$$

where we used that $\mathbb{P}_{\sigma^*,\tau^*}(m_{n-1} = \tilde{m}, \omega_n = \omega) = \lambda(\tilde{m})q_{\tilde{m}}(\omega)$.

Define

$$\overline{p}_m(\omega) := \mathbb{E}_{\sigma^*,\tau^*}\Big[\frac{1}{N\lambda(m)}\sum_{t=1}^{N}\mathbf{1}_{\{\omega_t=\omega,\theta_t=m\}}\Big].$$

That is, $\lambda(m)\overline{p}_m(\omega)$ is the expected fraction of periods in a block in which the state is ω and the message used is m.

For fixed N, since the strategy restarts every block, the discounted payoff converges to the average payoff within a block as $\delta \to 1$.

$$\lim_{\delta\to 1}\gamma_i^\delta(\sigma^*,\tau^*) = \sum_a \lambda(m)\sum_\omega \overline{p}_m(\omega)\sum_a \kappa(a \mid m)u_i(\omega, a) \quad \forall i \in \{S, R\}. \tag{A.1}$$

We will show that for sufficiently large block length, this payoff can be made arbitrarily close to the target payoff $e(\mu, \lambda)$.

First, observe that by a law of large numbers for Markov chains, if N is large enough, then with high probability the quota of each message is exhausted only in a small fraction of periods. Hence, under (σ^*, τ^*), the expected fraction of periods in which message and actions differ can be made arbitrarily small. It follows that $\overline{p}_m$ can be made arbitrarily close to p_m for every $m \in M$.

Using the fact that payoffs are bounded and the linearity of the expected payoff in RHS of (A.1), it follows that for every $\epsilon > 0$, there exists $N_0 \in \mathbb{N}$ and δ_0 such that for all $N \geq N_0$ and $\delta \geq \delta_0$, it holds that

$$\|\gamma^\delta(\sigma^*,\tau^*) - e(\mu,\lambda)\|_\infty \leq \epsilon.$$

□

We now show that the strategy profile (σ^*, τ^*) is an approximate equilibrium for sufficiently patient players.

Lemma 2. *For every $\epsilon > 0$, there exists $N_0 \in \mathbb{N}$ and $\delta_0 \in (0, 1)$ such that for all $N \geq N_0$ and $\delta \geq \delta_0$, the strategy profile (σ^*, τ^*) is an ϵ-equilibrium.*

Proof. Fix $\epsilon > 0$. We show that there exist N_0 and δ_0 such that for all $N \geq N_0$ and all $\delta \geq \delta_0$, no player can gain more than ϵ by deviating from (σ^*, τ^*).

Sender's deviations. Since play restarts at each block and continuation play is independent of past blocks, any long-run gain from deviation is bounded by the maximal gain from deviating within a single block. Fix any deviation $\tilde{\sigma}$. Since under τ^* the receiver plays according to message m exactly $N\lambda(m)$ times in each block, define

$$\tilde{p}_m(\omega) := \frac{1}{N\lambda(m)} \mathbb{E}_{\tilde{\sigma},\tau^*}\left[\sum_{n=1}^{N} \mathbf{1}_{\{\omega_n=\omega,\theta_n=m\}}\right].$$

Then $(\tilde{p}_m)_{m\in M}$ satisfies Bayes plausibility with weights λ, that is, $\mu = \sum_m \lambda(m)\tilde{p}_m$. So, the collection $(\tilde{p}_m)_{m\in M}$ is feasible and belongs to $\Sigma(\mu,\lambda)$.

For fixed N, as $\delta \to 1$, the sender's discounted payoff under $(\tilde{\sigma},\tau^*)$ converges to the average payoff within a block. Therefore, for δ sufficiently close to 1,

$$\gamma_S^\delta(\tilde{\sigma},\tau^*) \le \sum_{m\in M} \lambda(m) \sum_{\omega\in\Omega} \tilde{p}_m(\omega) \sum_{a\in A} \kappa(a \mid m) u_S(\omega,a) + \epsilon.$$

Since $(p_m)_{m\in M}$ has no profitable deviation within $\Sigma(\mu,\lambda)$, we have

$$\sum_{m\in M} \lambda(m) \sum_{\omega\in\Omega} \tilde{p}_m(\omega) \sum_{a\in A} \kappa(a \mid m) u_S(\omega,a) \le e_S(\mu,\lambda).$$

Combining the two inequalities gives

$$\gamma_S^\delta(\tilde{\sigma},\tau^*) \le e_S(\mu,\lambda) + \epsilon.$$

By Lemma 1, for N large enough and δ sufficiently close to 1,

$$\gamma_S^\delta(\sigma^*,\tau^*) \ge e_S(\mu,\lambda) - \epsilon.$$

Therefore,

$$\gamma_S^\delta(\tilde{\sigma},\tau^*) - \gamma_S^\delta(\sigma^*,\tau^*) \le 2\epsilon.$$

After replacing ϵ by $\epsilon/2$, it follows that the sender cannot gain more than ϵ by deviating.

Receiver's deviations. Because the receiver's actions are unobservable, his choice in period n does not affect the sender's future behavior. Hence, against σ^*, a myopic best response at every period is optimal. Let τ^{myp} denote the myopic best-response strategy in which, after every history, the receiver best responds to his current belief p_t.

It therefore suffices to show that deviating to τ^{myp} does not result in a gain greater than ϵ. By construction, τ^* plays a myopic best response to sender's message m whenever the quota for m is not exhausted. By the law of large numbers for Markov chains, for N large enough the empirical frequency of messages is close to λ with high probability. Hence, τ^* coincides with τ^{myp} in all but a small fraction of periods in each block. Since stage payoffs are bounded, the resulting difference in the discounted payoffs can be made smaller than ϵ by choosing N sufficiently large. Thus,

$$\gamma_R^\delta(\sigma^*,\tau^{\text{myp}}) - \gamma_R^\delta(\sigma^*,\tau^*) \le \epsilon.$$

Therefore, the receiver cannot gain more than ϵ by deviating. $\square$

Lemma 1 and Lemma 2 show that any $e \in \mathcal{E}(\mu, \lambda)$ is a uniform equilibrium payoff. We now show that players can obtain any convex combination of such payoffs.

Pick any $e \in \mathrm{Co}_\lambda\big(\mathcal{E}(\mu, \lambda)\big)$. Since payoffs lie in a finite-dimensional space, e can be written as a finite convex combination. Thus, there exist a finite index set I, weights $(\eta_i)_{i \in I}$ with $\eta_i \geq 0$ and $\sum_{i \in I} \eta_i = 1$, and marginals $\lambda_i \in \Delta(M)$ such that

$$e = \sum_{i \in I} \eta_i e(\mu, \lambda_i), \qquad \text{with } e(\mu, \lambda_i) \in \mathcal{E}(\mu, \lambda_i) \text{ for each } i \in I.$$

The players use a block strategy. However, each block of length N is further subdivided into I sub-blocks.[18] Sub-block i starts at period $N \sum_{j=1}^{i-1} \eta_j + 1$ and ends at period $N \sum_{j=1}^{i} \eta_j$. The players play according to strategy profile (σ_i^*, τ_i^*) in sub-block i.

Using the same argument as in Lemma 1, for each sub-block i, taking N sufficiently large ensures that the payoff induced by (σ_i^*, τ_i^*) is arbitrarily close to $e(\mu, \lambda_i)$ for all δ sufficiently close to 1. Since sub-block i occupies a fraction η_i of each block, it follows that

$$\lim_{\delta \to 1} \gamma^\delta(\tilde{\sigma}, \tilde{\tau}) = \sum_{i \in I} \eta_i e(\mu, \lambda_i).$$

It remains to check incentives. Because the strategy restarts at the beginning of each sub-block, deviations in one sub-block do not affect continuation play in subsequent sub-blocks. Thus, any unilateral deviation decomposes into deviations within sub-blocks.

By Lemma 2, for every $i \in I$, there exists δ_0^i such that for every $\delta \geq \delta_0^i$, no player can gain more than $\eta_i \epsilon$ by deviating in sub-block i. Let $\delta_0 = \max_{i \in I} \delta_0^i$. Then for every $\delta \geq \delta_0$, the total discounted gain from deviating is at most $\sum_{i \in I} \eta_i \epsilon = \epsilon$.

Using the convergence result, this implies that the difference in payoffs between the equilibrium strategy and any deviation is bounded above by 2ϵ. Therefore,

$$\sum_{i=1}^{I} \eta_i e(\mu, \lambda_i) \in \mathcal{U}.$$

A.2. **Proof of Proposition 1.** Consider the deviation τ^{myp} where the receiver plays a myopic best response to belief p_n at every history. Since continuation payoffs do not depend on the receiver's action, we have

(A.2)

$$\gamma_R(\sigma^*, \tau^{\mathrm{myp}}) - \gamma_R(\sigma^*, \tau^*) = \mathbb{E}_{\sigma^*, \tau^*}\left[\sum_{n=1}^{\infty} (1-\delta)\delta^{n-1}\Big(\mathbb{E}_{p_n}[u_R(\cdot, a^*(p_n))] - \mathbb{E}_{p_n}[u_R(\cdot, a_n)]\Big)\right].$$

Since (σ^*, τ^*) is an ϵ-equilibrium, the left-hand side is at most ϵ.

Fix any $\eta > 0$, and let

$$R_\eta^n := \{\mathbb{E}_{p_n}[u_R(\cdot, a^*(p_n))] - \mathbb{E}_{p_n}[u_R(\cdot, a_n)] \geq \eta\}.$$

[18] Choose N such that $N\eta_i\lambda_i(a)$ is an integer for all $i \in I$ and $a \in A$.

denote the event when the receiver's gain in period n is at least η.

Then (A.2) implies

$$\epsilon \geq \mathbb{E}_{\sigma^*,\tau^*}\left[\sum_{n=1}^{\infty}(1-\delta)\delta^{n-1}\Big(\mathbb{E}_{p_n}[u_R(\cdot,a^*(p_n))]-\mathbb{E}_{p_n}[u_R(\cdot,a_n)]\Big)\right] \geq \eta\,\mathbb{E}_{\sigma^*,\tau^*}\left[\sum_{n=1}^{\infty}(1-\delta)\delta^{n-1}\mathbf{1}_{R_\eta^n}\right].$$

Hence,

$$\mathbb{E}_{\sigma^*,\tau^*}\left[\sum_{n=1}^{\infty}(1-\delta)\delta^{n-1}\mathbf{1}_{R_\eta^n}\right] \leq \frac{\epsilon}{\eta}.$$

Therefore, for every $\eta > 0$, the discounted weight of periods in which the receiver's action is worse than a myopic best response by at least η vanishes as $\epsilon \to 0$.

Fix $\epsilon > 0$, and let (σ^*, τ^*) be an ϵ-equilibrium of the δ-discounted game for all $\delta \geq \delta_0$. Let

$$\nu(\omega, a) := (1-\delta)\sum_{n=1}^{\infty}\delta^{n-1}\mathbb{E}_{\sigma^*,\tau^*}[\mathbf{1}_{\{\omega_n=\omega,a_n=a\}}]$$

denote the discounted expected outcome induced by the strategy profile. Since the initial state is drawn from the invariant distribution μ, it follows that $\sum_a \nu(\omega, a) = \mu(\omega)$.

For the sender, we construct an undetectable deviation σ'. Since (σ^*, τ^*) is an ϵ-equilibrium, this deviation can improve the sender's payoff by at most ϵ. Using Lemma 4 in Renault, Solan, and Vieille (2013), in pseudo-renewal Markov chains, one can construct a sequence of fictitious states (ξ_n) that is statistically indistinguishable from the sequence (ω_n). In particular, (i) the sequence $\{\xi_n\}_n$ has the same law as $\{\omega_n\}_n$, (ii) in each period n, the law of the pair (ω_n, ξ_n) is given by distribution $c \in \mathcal{C}(\mu)$ and (iii) the conditional law of ω_n given $\xi_1, \ldots, \xi_n$ is given by $c(\cdot \mid \xi_n)$. Moreover, the sequence ξ_n can be constructed using information available only in period n. In any period n, the sender sends a message according to σ^* evaluated at the fictitious history $(\xi_1, m_1, \ldots, \xi_n)$. Formally, given a history $(\omega_1, \xi_1, m_1, \ldots, \omega_n, \xi_n)$ consisting of realized and fictitious states and messages, the strategy σ' selects the current message according to the distribution that σ^* would assign after the fictitious history $(\xi_1, m_1, \ldots, \xi_n)$.

We now show that the sender's expected payoff under the strategy profile (σ', τ^*) is equal to $\sum_{\omega,\xi,a} c(\omega \mid \xi)\nu(\xi, a)u_S(\omega, a)$.

$$\begin{aligned}
\mathbb{E}_{\sigma',\tau^*}[u_S(\omega_n, a_n)] &= \sum_{\omega}\sum_{a}\mathbb{P}_{\sigma',\tau^*}(\omega_n = \omega, a_n = a)u_S(\omega, a) \\
&= \sum_{\omega_n}\sum_{\xi_1,\ldots,\xi_n}\sum_{a_1,\ldots,a_n}\mathbb{P}_{\sigma',\tau}(\xi_1, a_1, \ldots, \omega_n, \xi_n, a_n)u_S(\omega_n, a_n) \\
&= \sum_{\omega_n}\sum_{\xi_1,\ldots,\xi_n}\sum_{a_1,\ldots,a_n}\mathbb{P}_{\sigma',\tau}(\omega_n \mid \xi_1, \ldots, \xi_n)\mathbb{P}_{\sigma',\tau}(\xi_1, a_1, \ldots, \xi_n, a_n)u_S(\omega_n, a_n) \\
&= \sum_{\omega_n}\sum_{\xi_n}\sum_{a_n}c(\omega_n \mid \xi_n)\mathbb{P}_{\sigma',\tau}(\xi_n, a_n)u_S(\omega_n, a_n)
\end{aligned}$$

Since the process (ξ_n) has the same law as (ω_n), and since σ' applies σ^* to the fictitious history, the discounted joint distribution over (ξ_n, a_n) under (σ', τ^*) coincides with the discounted joint distribution of (ω_n, a_n) under (σ^*, τ^*). Hence,

$$(1-\delta)\sum_{n=1}^{\infty}\delta^{n-1}\mathbb{P}_{\sigma',\tau^*}(\xi_n = \xi, a_n = a) = \nu(\xi, a).$$

Therefore,

$$\gamma_S^{\delta}(\sigma', \tau^*) = \sum_{\omega,\xi,a} c(\omega \mid \xi)\nu(\xi, a)u_S(\omega, a).$$

So, for any outcome to be an equilibrium, there must be no profitable deviation in the set of deviations defined by $\mathcal{M}(\nu)$.

A.3. **Proof of Proposition 2.** The inclusion follows from Theorem 1. We prove the converse.

Fix $\epsilon > 0$, and let (σ, τ) be an ϵ-equilibrium of the δ-discounted game for all $\delta \geq \delta_0$.

The receiver part follows as in Proposition 1. Since continuation payoffs do not depend on the receiver's current action, a myopic best response is optimal. Hence, for every $\eta > 0$, the discounted weight of periods in which the receiver's action is more than η below a myopic best response is bounded by ϵ/η, and thus vanishes as $\epsilon \to 0$.

Consider the sender's deviations. At any history $h_n = (m_1, \ldots, m_n)$, the sender's strategy induces a marginal distribution $\lambda_{h_n} \in \Delta(A)$ over action recommendations. Since the state process is i.i.d., the receiver's belief before observing the message is always equal to the prior μ.

Consider the deviation σ', where at any history h_n, if there exists a profitable deviation in $\Sigma(\mu, \lambda_{h_n})$, the sender switches to one that achieves the highest profit, and otherwise follows σ. Such a deviation is undetectable because it keeps the distribution over the message process unchanged. Also, from the i.i.d assumption, it does not affect continuation payoffs since future states are independent of the current one.

Like before, fix any $\eta > 0$ and let S_η^n be the event that in period n there exists a feasible one-shot deviation in $\Sigma(\mu, \lambda_n)$ that improves the sender's current payoff by at least η.

We have

$$\gamma_S(\sigma', \tau) - \gamma_S(\sigma, \tau) \geq \eta \cdot \mathbb{E}_{\sigma,\tau}\Big[\sum_{n=1}^{\infty}(1-\delta)\delta^{n-1}\mathbf{1}_{S_\eta^n}\Big]$$

Since (σ, τ) is an ϵ-equilibrium, it follows that

$$\mathbb{E}_{\sigma,\tau}\Big[\sum_{n=1}^{\infty}(1-\delta)\delta^{n-1}\mathbf{1}_{S_\eta^n}\Big] \leq \frac{\epsilon}{\eta}.$$

Since this holds for every $\eta > 0$, the discounted weight of periods in which the sender has a deviation gain bounded away from zero vanishes as $\epsilon \to 0$. It follows that the sender's static incentive constraint holds in the limit.

Thus, for any $\eta > 0$, the discounted weight of periods in which either the sender's or the receiver's equilibrium condition is violated by at least η is at most $2\epsilon/\eta$. Hence, in the limit, the contribution of such periods to the discounted payoff vanishes.

It follows that any limit point of discounted equilibrium payoffs can be approximated by a convex combination of stage outcomes that satisfy obedience and the sender's incentive constraint for some marginal distribution λ. Therefore, any limit point belongs to $\mathrm{Co}_\lambda\big(\mathcal{E}(\mu, \lambda)\big)$.

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